

While Loop, For loop and Functions

Overview:

This documentation presents three Python programs that demonstrate fundamental programming concepts. The programs include a Number Guessing Game using a while loop, a Multiplication Table using a for loop, and a BMI Calculator using a user-defined function. These programs demonstrate concepts such as loops, conditional statements, user input, type conversion, arithmetic operations, break, continue, range(), random number generation, functions, parameters, arguments, and return statements.

1. Number Guessing Game(While loop):

This program creates a simple number guessing game using a while loop. First, the random module is imported and the randint() function generates a random number between 1 and 10, which is stored as the secret number. The user is given three attempts to guess the number. A while loop repeats the guessing process, while if, elif, and else statements compare the user's guess with the secret number. If the user enters a number outside the range of 1 to 10, an invalid input message is displayed, and the continue statement skips the current iteration. If the guess is greater than the secret number, the program displays "your guess is too high", and if it is smaller, it displays "your guess is too low." When the user guesses the correct number, the program displays "your guess is correct" and the break statement stops the loop. The number of attempts is reduced after every valid incorrect guess. This program demonstrates random numbers, while loops, conditional statements, break, continue, and user input.

2. Multiplication Table (For loop):

This program generates the multiplication table of a number entered by the user. First, the program uses input() to get a number from the user and int() converts the input into an integer. A for loop with range(1, 11) is then used to repeat the multiplication from 1 to 10. During each iteration, the value of i represents the multiplier, and num * i calculates the result. The program displays each multiplication step in a clear format. This program helps in understanding the use of user input, for loops, range(), variables, and arithmetic operators in Python.

3. BMI Calculator Using a Function:

This program calculates the Body Mass Index (BMI) using a user-defined function called calculate_bmi. The function accepts two parameters, weight and height, and calculates BMI using the formula $BMI = Weight / Height^2$. The calculated value is returned using the return statement. The program then prompts the user to enter their weight in kilograms and height in meters. The float() function converts the input values into decimal numbers. Finally, the calculate_bmi() function is called by passing the user's weight and height as arguments, and the calculated BMI is displayed. This program demonstrates user-defined functions, parameters, arguments, return statements, user input, type conversion, and arithmetic operations.

Conclusion:

These three programs demonstrate important Python programming concepts through simple practical examples. The Number Guessing Game demonstrates the use of a while loop, conditional statements, break, continue, user input, and random number generation. The Multiplication Table demonstrates the use of a for loop, range(), variables, user input, type conversion, and arithmetic operations. The BMI Calculator demonstrates the use of a user-defined function, parameters, arguments, a return statement, type conversion, and arithmetic operations. Together, these programs provide a basic understanding of loops and functions in Python.

Outputs:

```
Enter the number:5
your guess is too high,try again
Enter the number:11
Invalid input range,Give input range between 1 to 10
Enter the number:1
your guess is too low,try again
Enter the number:3
your guess is too high,try again
Enter the number:2
wowieee!!!your guess is correct
```

```
... Enter the number:5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

```
Enter your weight in kg: 74
Enter your height in meters: 1.75
Your BMI is: 24.2
```